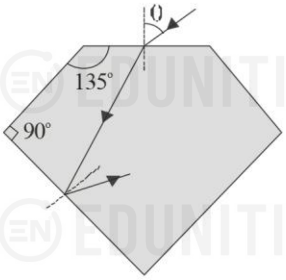
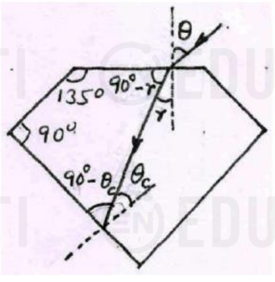
1. A ray of light enters a diamond ($\mu = 2$) from air and is being internally reflected near the bottom as shown in the figure. Find maximum value of angle θ possible.



Solution.



The critical angle for diamond ($\mu = 2$) is

$$\theta_c = \sin^{-1}(1/\mu) = \sin^{-1}(1/2) = 30^\circ$$

From figure,

$$(90^\circ - r) + (90^\circ - \theta_c) + 90^\circ + 135^\circ = 360^\circ$$

$$\Rightarrow r = 45^\circ - \theta_c = 45^\circ - 30^\circ$$

Now, $\mu = \dfrac{\sin\theta}{\sin r}$

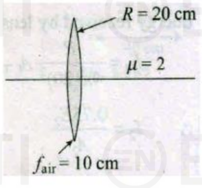
$$\Rightarrow \sin\theta = \mu \sin r = 2\sin(45^\circ - 30^\circ)$$

$$= 2(\sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ)$$

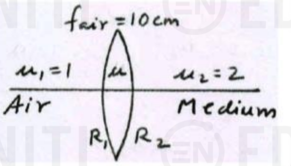
$$= 2\left(\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}\right) = \frac{\sqrt{3}-1}{\sqrt{2}}$$

$$\therefore \boxed{\theta = \sin^{-1}\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)}$$

2. Focal length of a thin lens in air is 10 cm. Now, medium on one side of the lens is replaced by a medium of refractive index 2. The radius of curvature of the surface of lens, in contact with the medicem, is 20 cm. Find the new focal length of the lens when the object is placed in air on the other side.



Solution Let R_1 and R_2 be the radii of the left and right surfaces. Let μ be the refractive index of the lens and f_{air} be its focal length in air.



Then $f_{air} = 10$ cm and $R_2 = -20$ cm

To find the new focal length f, we shall consider the object to be placed in air at infinity so that the final image is formed at distance f. If the image formed by the the surafce is at distance v, then for the left and right surfaces, we have

$$\frac{\mu}{v} - \frac{1}{\infty} = \frac{\mu - 1}{R_1} \quad \text{and} \quad \frac{2}{f} - \frac{\mu}{v} = \frac{2-\mu}{R_2}$$

On adding the two equations, we get

$$\frac{2}{f} = \frac{\mu-1}{R_1} + \frac{2-\mu}{R_2} = \frac{\mu-1}{R_1} + \frac{1-\mu}{R_2} + \frac{1}{R_2}$$

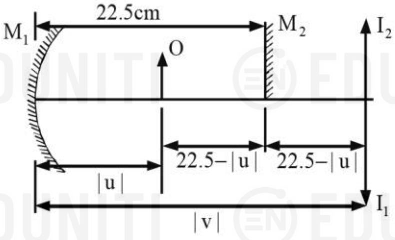
$$= (\mu-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) + \frac{1}{R_2} = \frac{1}{f_{air}} + \frac{1}{R_2} = \frac{1}{10} + \frac{1}{-20} = \frac{1}{20}$$

$$\therefore \boxed{f = 40\,\text{cm}}$$

3. A plane mirror is placed 22.5 cm in front of a concave mirror of focal length 10 cm. Find, where an object can be placed between the two mirrors, so that the first image in both mirrors coincide.

Solution: Let O be the object.

The image formed by the plane mirror M_2 is I_2, and by the concave mirror M_1 should then form a real inverted image I_1 so that the two images coincide.



Now, $|v| = 22.5 + (22.5 - |u|) = 45 - |u|$ or $-v = 45 + u$

Here, $f = -10$ cm

Also, $\dfrac{1}{f} = \dfrac{1}{v} + \dfrac{1}{u}$ $\quad \Rightarrow \dfrac{1}{-10} = \dfrac{1}{-(45+u)} + \dfrac{1}{u}$

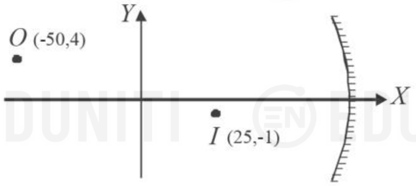
$\Rightarrow u^2 + 45u + 450 = 0$ $\quad \Rightarrow (u+30)(u+15) = 0$

$\Rightarrow u = -30$cm or $u = -15$cm,

$u = -30$cm is not admissible. Hence, $u = -15$cm

Therefore, the object must be placed at a distance of 15cm from the concave mirror.

4. A concave mirror forms an image I corresponding to a point object O. Find the equation of the circle intercepted by the xy plane on the mirror is



Solution Let the coordinates of the pole of the mirror be $(x_0, 0)$

Then, $u = -(x_0 + 50)$ and $v = -(x_0 - 25)$

The heights of object and image are respectively

$h_0 = 4$ and $h_i = -1$

Now, $m = -\dfrac{v}{u} = \dfrac{h_i}{h_0}$ $\Rightarrow -\dfrac{-(x_0-25)}{-(x_0+50)} = \dfrac{-1}{4}$

$\Rightarrow 4(x_0 - 25) = x_0 + 50$ $\Rightarrow x_0 = 50$

$\Rightarrow u = -(x_0 + 50) = -100$ and $v = -(x_0 - 25) = -25$

From mirror equation,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} = \frac{1}{-100} + \frac{1}{-25} = \frac{-5}{100}$$

$\Rightarrow f = -20$ and $R = 2|f| = 40$

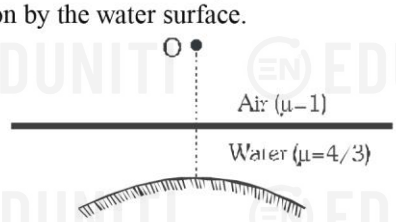
The centre of the required circle lies on x -axis at

$x_0 - R = 50 - 40 = 10$

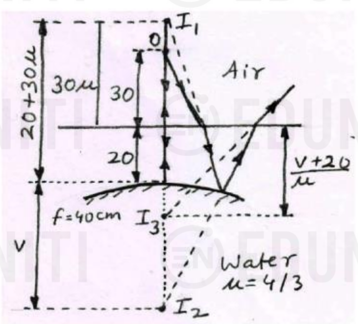
Therefore, the equation of the circle is

$(x-10)^2 + (y-0)^2 = 40^2$ or $\boxed{x^2 + y^2 - 20x - 1500 = 0}$

5. A point object lies 30 cm above water on the axis of a convex mirror of focal length 40 cm lying 20 cm below water surface. Find the location of final image formed after refraction by water surface followed by reflection from mirror and again refraction by the water surface.



Solution.



The image I_1 of object O formed due to refraction by water surface is $30\mu = 30 \times 4/3 = 40\,\text{cm}$ above the water surface.

It is at a distance $20 + 30\mu = 60\,\text{cm}$ from the convex mirror of focal lenth $f = 40$ cm. The image I_2 formed by reflection is at a distance v from the mirror given by

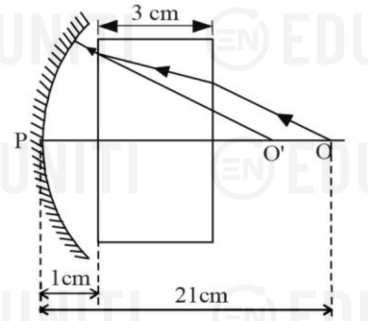
$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{40} - \frac{1}{-60} = \frac{1}{24} \qquad \Rightarrow v = 24\,\text{cm}$$

It is at a depth $v + 20 = 44$ cm below the surface of water.

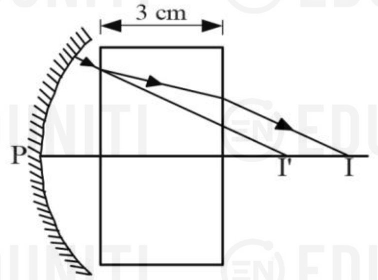
The final image I_3 formed due to refraction again by water surface is at a depth

$$\frac{v+20}{\mu} = \frac{24+20}{4/3} = \boxed{33\,\text{cm}} \text{ below the water surface.}$$

6. An object is placed 21 cm in front of a concave mirror of radius of curvature 10 cm. A glass plate of thickness 3 cm and refractive index 1.5 is then placed close to the mirror in the space between the object and the mirror. Find the position of the final image formed. (You may take the distances of the nearer surface of the plate from the mirror to be 1 cm).



Solution:



The rays starting from the object O would appear to come from the point O', due to refraction through the glass plate.

This displacement OO' is given by

$$OO' = t\left(1 - \frac{1}{\mu}\right) = 3\left(1 - \frac{1}{1.5}\right) = 1\,\text{cm}$$

$\Rightarrow PO' = 21 - 1 = 20\,\text{cm}$

Now if the object were at O' (apparent position), its image formed by the mirror would have been at a distance v from the mirror (at I') so that

$\dfrac{1}{f} = \dfrac{1}{v} + \dfrac{1}{u}$ where, $u = -20\,\text{cm}$, $f = \dfrac{R}{2} = -5\,\text{cm}$

$$\Rightarrow v = \frac{uf}{u-f} = \frac{(-20)\times(-5)}{(-20)-(-5)} = -6.7\,\text{cm}$$

$\Rightarrow PI' = -v = 6.7\,\text{cm}$

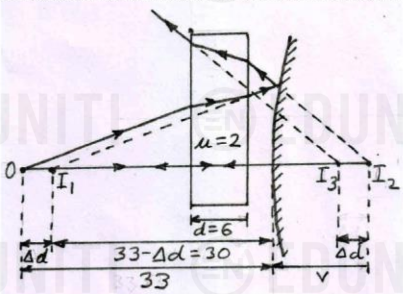
However, the reflected rays have to pass through the glass slab and would converge at I.

But $II' = 3\left(1 - \frac{1}{1.5}\right) = 1\text{cm}$

Therefore, the final position of the image will be at I at a distance $PI = 6.7 + 1 = 7.7\text{cm}$ from the mirror.

7. A point object is placed 33 cm from a convex mirror of radius of curvature 40 cm. A glass plate of thickness 6 cm and refractive index 2.0 is placed between the object and mirror, close to the mirror. Find the distance of final image from the object.

Solution.



The image I_1 of object O due to glass slab is shifted by

$$\Delta d = d\left(1 - \frac{1}{\mu}\right) = 6\left(1 - \frac{1}{2}\right) = 3\,\text{cm towards right.}$$

I_1 acts as an object for the mirror. Now, for the mirror,

$u = -(33-3) = -30$ cm and $f = R/2 = 40/2 = 20$ cm

$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} - \frac{1}{-30} = \frac{1}{12} \qquad \Rightarrow v = 12\text{cm}$$

The image I_2 formed by the mirror is at a distance 12 cm from the mirror as shown in figure.

The final image I_3 due to glass slab is shifted by

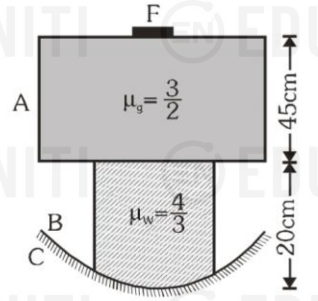
$\Delta d = 3\text{cm}$

towards left.

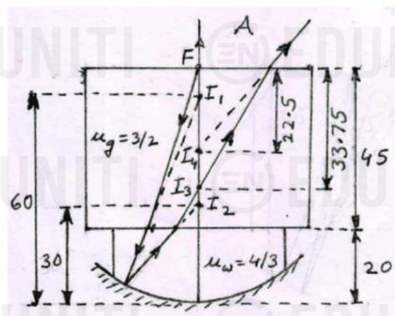
Therefore, the distance of the final image I_3 from the object O is

$$33 + v - \Delta d = 33 + 12 - 3 = \boxed{42\,\text{cm}}$$

8. A fly F is sitting on a glass slab A, 45 cm thick and of refractive index 3/2. The slab covers the top of a container B containing water (refractive index 4/3) upto a height of 20 cm. The bottom of container is closed by a concave mirror C of radius of curvature 40 cm. Locate the final image formed by all refractions and reflection assuming paraxial rays.



Solution. We shall consider water as the reference medium since water is in contact with the mirror and the mirror equation is independent of the medium.



The image I_1 of fly F due to the glass slab is shifted by

$$45\left(1 - \frac{\mu_\omega}{\mu_g}\right) = 45\left(1 - \frac{4/3}{3/2}\right) = 45\left(1 - \frac{8}{9}\right) = 5\text{cm}$$

So, the distance of I_1 is $20 + 45 - 5 = 60$ cm from the mirror which acts as the object for the mirror.

Now, for the mirror,

$u = -60$ cm and $f = -R/2 = -40/2 = -20$ cm

$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-60} = -\frac{1}{30} \qquad \Rightarrow v = -30\text{cm}$$

The image I_2 formed by the mirror is 10 cm above the surface of water. Its image I_3 in glass slab is at a height

$$\frac{\mu_g}{\mu_\omega} \times 10 = \frac{3/2}{4/3} \times 10 = \frac{9}{8} \times 10 = 11.25\text{cm}$$

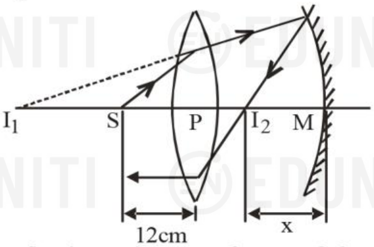
from the water surface and at a depth

$45 - 11.25 = 33.75$ cm from the top surface of glass slab.

The final image I_4 is at a depth

$$\frac{33.75}{\mu_g} = \frac{33.75}{1.5} = \boxed{22.5\text{cm}} \text{ from } F.$$

9. A converging lens of focal length 15 cm and a converging mirror of focal length 20 cm are placed with their principal axes coinciding. A point source S is placed on the principal axis at a distance of 12 cm from the lens as shown in figure. It is found that the final beam comes out parallel to the principal axis. Find the separation between the mirror and the lens.



Solution: Let us first locate image I of source S formed by the lens.

Here, $u=-12\text{cm}$ and $f=15\text{cm}$.

$$\Rightarrow \frac{1}{v}=\frac{1}{f}+\frac{1}{u}=\frac{1}{15}-\frac{1}{12}$$

$$\Rightarrow v=-60\text{cm} \qquad \therefore PI_1=60\text{cm}$$

The image I_1 acts as the object for the mirror. The mirror forms an image I_2 of the object I_1. This image I_2 then acts as the object for the lens and the final beam comes out parallel to the principal axis. Clearly, I_2 must be at the focus of the lens. We have,

$$I_1I_2=I_1P+PI_2=60\text{cm}+15\text{cm}=75\text{cm}$$

Suppose, the distance of the mirror from I_2 is x cm. For the reflection from the mirror,

$$u=MI_1=-(75+x)\text{cm}, \quad v=-x\text{ cm} \quad \text{and} \quad f=-20\text{cm}$$

Using $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, we get $\frac{1}{x}+\frac{1}{75+x}=\frac{1}{20}$

$$\Rightarrow \frac{75+2x}{(75+x)x}=\frac{1}{20}$$

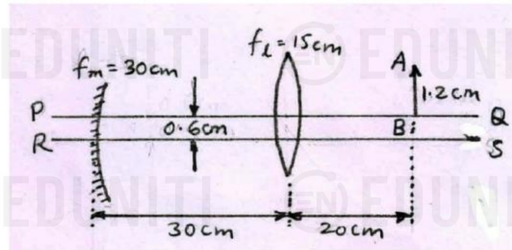
$$\Rightarrow x^2+35x-1500=0 \qquad \Rightarrow (x+60)(x-25)=0$$

This gives $x=25$ or -60

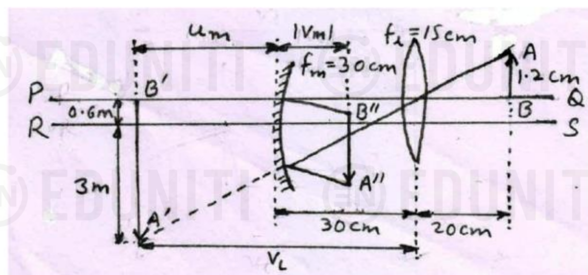
As the negative sign has no physical meaning, only positive sign should be taken. Taking $x=25$, the separation between the lens and the mirror is

$(15+25)$ cm=40 cm.

10. A convex lens of focal length 15 cm and a concave mirror of focal length 30 cm are kept with their optic axes PQ and RS parallel but separated in vertical direction by 0.6 cm as shown in fig. The distance between the lens and mirror is 30 cm. An upright object AB of height 1.2 cm is placed on the optic axis PQ of the lens at a distance of 20 cm from the lens. If $A''B''$ is the image after refraction from the lens and reflection from the mirror, find its distance from the pole of the mirror and obtain its magnification. Also, locate positions of A'' and B'' with respect to the optic axis RS.



Solution



For the lens, $u_l=-20$ cm and $f_l=15$ cm

$$\Rightarrow \frac{1}{v_l}=\frac{1}{f_l}+\frac{1}{u_l}=\frac{1}{15}+\frac{1}{-20}=\frac{1}{60} \qquad \Rightarrow v_l=60\text{cm}$$

$$\Rightarrow m_l=\frac{v_l}{u_l}=\frac{60}{-20}=-3$$

The size of image $A'B'$ is $|m_l|\times1.2=3.6\text{m}$ with end B' at 0.6m above RS and end A' at 3m below RS.

For the mirror, $f_m=30\text{cm}$ and $u_m=v_l-30=60-30=30\text{cm}$

$$\Rightarrow \frac{1}{v_m}=\frac{1}{f_m}-\frac{1}{u_m}=\frac{1}{-30}-\frac{1}{30}=-\frac{2}{30} \qquad \therefore \boxed{v_m=-15\text{cm}}$$

The location of final image $A''B''$ is 15 cm to the right if mirror.

$$m_m=-\frac{v_m}{u_m}=-\frac{-15}{30}=\frac{1}{2}$$

The size of image $A''B''$ is $m_m\times3.6=\boxed{1.8\text{m}}$

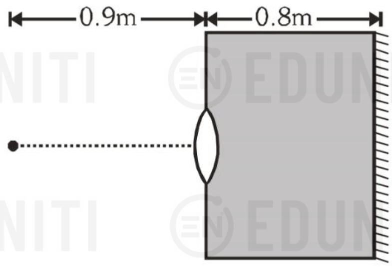
with end B'' at $0.6m_m=0.6\times1/2=0.3\text{m}$ above RS and end A'' at

$3m_m=3\times1/2=1.5\text{m}$ below RS.

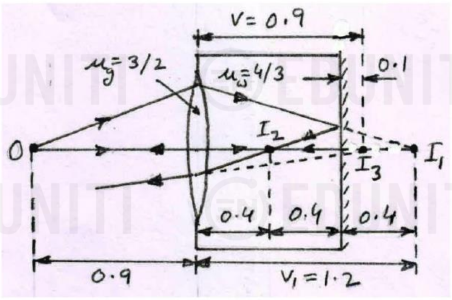
The magnification of final image $A''B''$ is

$$m=m_lm_m=-3\times1/2=\boxed{-1.5}$$

11. A thin equiconvex lens of glass of refractive index $\mu = 3/2$ and of focal length 0.3 m in air is sealed into an opening at one end of a tank filled with water $\mu = 4/3$. On the opposite side of the lens, a mirror is placed inside the tank on the tank wall perpendicular to the lens axis, as shown in figure. The separation between the lens and the mirror is 0.8 m. A small object is placed outside the tank in front of the lens at a distance of 0.9 m from the lens on its axis. Find the position (relative to the lens) of the image of the object formed by the system.



Solution.



If R is the radius of equiconvex lens, then

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = \frac{2(\mu - 1)}{R}$$

$$\Rightarrow R = 2(\mu - 1)f = 2(1.5 - 1) \times 0.3 = 0.3\,\text{m}$$

The image I_1 (in water) of the object O (in air) formed by the lens is at a distance v_1 from the lens where

$$\frac{4/3}{v_1} - \frac{1}{-0.9} = \frac{\mu_g - 1}{R} + \frac{\mu_w - \mu_g}{-R}$$

$$= \frac{3/2 - 1}{0.3} + \frac{4/3 - 3/2}{-0.3} = \frac{5}{3} + \frac{5}{9} = \frac{20}{9}$$

$$\Rightarrow \frac{4}{3v_1} = \frac{20}{9} - \frac{10}{9} = \frac{10}{9} \qquad \Rightarrow v_1 = 1.2\,\text{m}$$

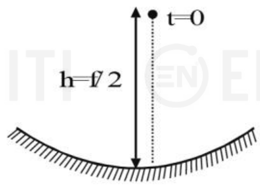
The image I_1 is 1.2 m to the right of lens and $1.2 - 0.8 = 0.4$ m to the right of mirror. Its image I_2 is formed at a distance 0.4 m to the left of mirror and $0.8 - 0.4 = 0.4$ m to the right of lens. The final image I_3 (in air) formed by the lens is at a distance v from the lens where

$$\frac{1}{v} - \frac{4/3}{-0.4} = \frac{\mu_g - \mu_w}{R} + \frac{1 - \mu_g}{-R} = \frac{20}{9}$$

$$\Rightarrow \frac{1}{v} = \frac{20}{9} - \frac{10}{3} = -\frac{10}{9} \qquad \therefore \boxed{v = -0.9\,\text{m}}$$

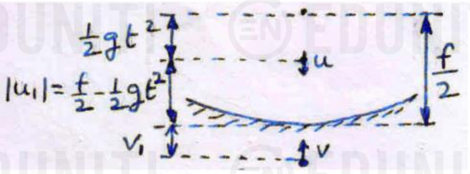
The final image I_3 is formed 0.9 m to the right of lens and $0.9 - 0.8 = 0.1$ m to the right of mirror.

12.



A particle is dropped along the axis from a height $h = f/2$ on a concave mirror of focal length f as shown in figure. Find the maximum speed of image.

Solution



After time t, the speed of the particle is $u = gt$ and its distance from the mirror is

$$|u_1| = \frac{f}{2} - \frac{1}{2}gt^2$$

To find the distance v_1 of image, we use mirror formula

$$\frac{1}{-(f/2 - gt^2/2)} + \frac{1}{v_1} = \frac{1}{-f}$$

$$\Rightarrow \frac{1}{v_1} = \frac{1}{f/2 - gt^2/2} - \frac{1}{f} = \frac{f/2 + gt^2/2}{f(f/2 - gt^2/2)}$$

The magnitication at this instant is

$$m = \frac{v_1}{f/2 - gt^2/2} = \frac{f}{f/2 + gt^2/2} = \frac{2f}{f + gt^2}$$

The speed of image is

$$v = um^2 = gt\left(\frac{2f}{f + gt^2}\right)^2 = \frac{4gf^2 t}{(f + gt^2)^2}$$

For maximim speed, $dv/dt = 0$

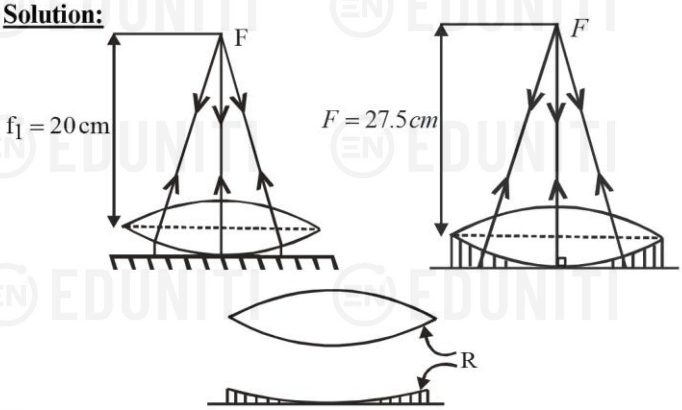
$$\Rightarrow (f + gt^2)^2 - t \times 2 \times 2gt(f + gt^2) = 0$$

$$\Rightarrow (f + gt^2) - 4gt^2 = 0$$

$$\Rightarrow t = \sqrt{\frac{f}{3g}} \qquad \therefore v_{\max} = \frac{4gf^2\sqrt{f/3g}}{(f + g \times f/3g)^2} = \boxed{\frac{3}{4}\sqrt{3gf}}$$

13. A thin equiconvex lens is placed on a horizontal plane mirror and a pin held 20cm above the lens coincides in position with its own image. The space between the lens and mirror is filled with water $(\mu = 4/3)$ and then to coincide with its own image as before, the pin has to be raised until its distance from the lens is 27.5cm. Find the radius of curvature of lens

Solution:



In the first case, the rays after refraction by lens, must fall normal to the mirror, i.e., must emerge parallel to the principle axis. This means that the object is kept at the focus of the lens.

$\Rightarrow f_1 = 20\,\text{cm}$

In the second case, the pin has to be placed at the focal length F of the combination of the convex lens and plano-concave lens made of water.

$\Rightarrow F = 27.5$ cm

If f_2 is the focal length of water lens, then

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \quad \Rightarrow \frac{1}{27.5} = \frac{1}{20} + \frac{1}{f_2} \quad \Rightarrow f_2 = -\frac{220}{3}\text{ cm}$$

For the water lens,

$$\frac{1}{f_2} = (\mu_w - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow -\frac{3}{220} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{R} - \frac{1}{\infty}\right) \qquad \Rightarrow R = -\frac{220}{9}\text{cm}$$

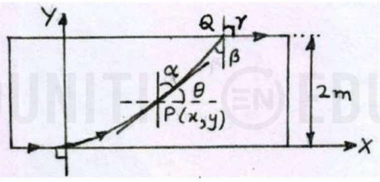
Therefore, the radius of curvature of the convex lens is

$R = 220/9\,\text{cm}$

14. A ray of light moving along x-axis is incident on a glass slab at origin at grazing incidence. The refractive index of the material of the slab is given by $\mu = \sqrt{1+y}$. If the thickness of the slab is $d = 2$ m, determine

(a) the equation of trajectory of the ray inside the slab
(b) the coordinates of point where the ray exits from the slab.
(c) the angle the ray makes with normal at the point of exit.

Solution The refractive index of the glass slab $\mu = \sqrt{1+y}$ is equal to 1 at the origin and keeps increasing along y-axis. As a result, the ray incident at origin along x-axis keeps bending towards the normal as y increases.



(a) Consider a general point $P(x, y)$ on its path. Let the tangent to the ray at P make angle α with vertical (normal) and θ with horizontal. The slope of the tangent at P is

$$m = \frac{dy}{dx} = \tan\theta = \cot\alpha$$

At at all points, $\mu \sin\alpha = \text{constant}$. At origin, $\mu = \sqrt{1+0} = 1$ and $\alpha = 90^\circ$

$\Rightarrow \mu\sin\alpha = 1 \times 90^\circ = 1$ or $\sin\alpha = 1/\mu$

$$\Rightarrow \frac{dy}{dx} = \cot\alpha = \frac{\sqrt{1-\sin^2\alpha}}{\sin\alpha} = \frac{\sqrt{1-1/\mu^2}}{1/\mu}$$

$$= \sqrt{\mu^2 - 1} = \sqrt{(1+y)-1} = \sqrt{y}$$

$$\Rightarrow \int_0^y \frac{dy}{\sqrt{y}} = \int_0^x dx \qquad \Rightarrow 2\sqrt{y} = x.$$

$\therefore \boxed{y = x^2/4}$

is the equation of the trajectory of the ray inside the slab.

(b) At point Q (where the ray exits the slab),

$y = 2\text{m}$ and $x = 2\sqrt{y} = 2\sqrt{2}\text{ m}$

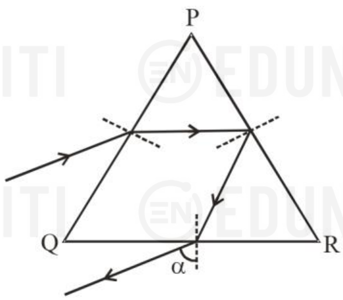
Therefore coordinates of Q are $\boxed{(2\sqrt{2}\text{ m}, 2\text{ m})}$

(c) At Q, inside the slab, $\mu = \sqrt{1+2} = \sqrt{3}$ and outside the slab, $\mu = 1$. As $\mu\sin\alpha = 1$,

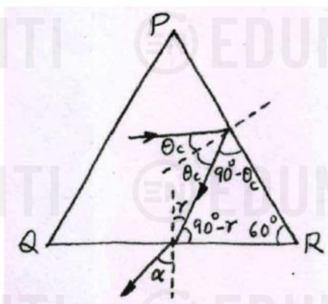
inside the slab, $\sqrt{3}\sin\beta = 1$ $\quad\therefore \boxed{\beta = \sin^{-1} 1/\sqrt{3}}$

and outside the slab, $1 \times \sin\gamma = 1$ $\quad\therefore \boxed{\gamma = 90^\circ}$

15. The figure shows the path of a ray passing through an equilateral prism PQR. It is incident on face PR at an angle slightly greater than the critical angle for total internal reflection. If the angle α shown in the figure is 30°, calculate the refractive index of the material of the prism.



Solution.



From figure,

$$(90° - \theta_c) + (90° - r) + 60° = 180° \qquad \Rightarrow r = 60° - \theta_c$$

$$\mu = \frac{\sin\alpha}{\sin r} = \frac{\sin 30°}{\sin(60° - \theta_c)}$$

$$\Rightarrow \frac{1}{2\mu} = \sin(60° - \theta_c)$$

$$= \sin 60°\sqrt{1 - \sin^2\theta_c} - \cos 60° \sin\theta_c$$

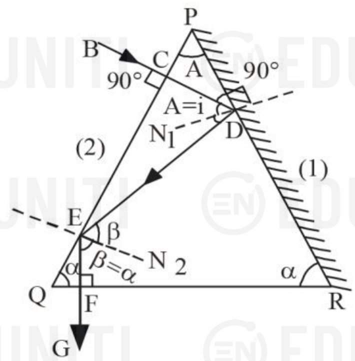
$$= \frac{\sqrt{3}}{2}\sqrt{1 - \frac{1}{\mu^2}} - \frac{1}{2} \times \frac{1}{\mu}$$

$$\Rightarrow 1 = \sqrt{3(\mu^2 - 1)} - 1 \qquad \Rightarrow 3(\mu^2 - 1) = 4$$

$$\therefore \boxed{\mu = \sqrt{7/3}}$$

16. The cross-section of a glass prism has the form of an isosceles triangle. One of the refracting faces is silvered. A ray of light falling normally on the other refracting face, reflecting twice emerges through the base of the prism perpendicular to it. Find the angles of the prism.

Solution:



The incident ray BC at normal incidence is refracted along CD, suffers reflection at silvered face, along DE and at E it again suffers total internal reflection along EF which is normal to the base and finally emerges along FG. From the figure, we can see that

$i = A$, Also, $\beta = \alpha$

Since, $EN_2 \parallel CD$, $\beta = 2i$ $\Rightarrow \alpha = 2A$

In ΔPQR, $2\alpha + A = 180°$

On solving, we get, $A = 36°$ and $\alpha = 72°$